

1. Administrative & Study Identification

Full Study Title: A Randomized, Double-Blind Study to Evaluate the Safety, Tolerability, and Immunogenicity of V118 Formulation E in Healthy Adults **Short Title/Acronym:** A Study of V118E in Healthy Participants (V118E-003) **Protocol Number:** V118E-003 **NCT Number:** NCT07168915 **Posting ID:** 311987672409458874 **Sponsor/Company Name:** Merck **Investigational Product:** V118E (Investigational Pneumococcal Vaccine) **Phase of Development:** Phase 1 **Trial Status:** RECRUITING (Currently recruiting participants) **Medical Monitor:** Clinical Director, Merck

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the safety and tolerability of the V118 formulation E vaccine in healthy adults. **Secondary Objectives:** To evaluate the immune response (immunogenicity) generated by the V118E vaccine by measuring antibody levels against specific pneumococcal serotypes.

2.2 Background & Rationale To address the ongoing global health concern of pneumococcal disease, which can cause severe illnesses ranging from ear infections and sinusitis to pneumonia, bloodstream infections, and meningitis. As bacterial strains evolve, new vaccines are needed. V118E is a newly formulated vaccine tested in a 2-dose regimen, aiming to induce a stronger and potentially longer-lasting immune response than existing single-dose standard vaccines (like PREVNAR 20™) to better prevent *Streptococcus pneumoniae* infections.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Active Comparator (PREVNAR 20™) and Placebo-controlled (Sodium Chloride / Saline) **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration **Target N\$:** 126 healthy participants **Number of Sites:** Multiple US sites (e.g., Florida, Missouri, Texas) **Estimated Study Start:** October 2025 **Estimated Primary Completion:** November 2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy adults aged 18 to 65 years. 2. Is in good health before randomization. 3. Ability to understand the study and provide informed consent.

4.2 Exclusion Criteria 1. Has a history of invasive pneumococcal disease or known history of other culture-positive pneumococcal disease within 3 years prior to receiving study vaccination. 2. Has a history of clinically significant endocrine, gastrointestinal (GI), cardiovascular, hematological, hepatic, immunological, renal, respiratory, genitourinary, or major neurological (including stroke and chronic seizures) abnormalities or diseases. 3. Prior receipt of experimental pneumococcal vaccines or concurrent participation in another clinical trial.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: 2-dose regimen administered on Day 1 and Day 29. **Route of Administration:** Intramuscular (IM) injection. **Duration of Treatment:** 29 days of active vaccination dosing followed by a safety and immunogenicity observation period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard routine medications deemed safe and non-interfering by the investigator. **Prohibited:** Immunosuppressive treatments or immunomodulating therapies that may skew the immunogenicity assessment.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	-14 to 0	Screening, Informed Consent, Medical History, Baseline Health Assessment
2	0 to 1	Randomization, First IM Injection (V118E or PREVNAR 20™), Dispense Vaccine Report Card (VRC)
3	29	Interim, Second IM Injection (V118E or Sodium Chloride Placebo), Safety Check, VRC Review
4	Follow-up	End of Study, Final Safety Assessment, Collection of VRCs, Immunogenicity Blood Draws (Antibody Titers)

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes Definition: Safety and tolerability profile, determined by the incidence of adverse events (AEs). **Specific Measurements:** 1. **Number of Participants With Solicited Injection-Site Adverse Events (AEs):** Including redness, swelling, and pain or tenderness recorded on the participant's vaccine report card (VRC). 2. **Number of Participants With Solicited Systemic AEs:**

Including fever, muscle aches all over the body, joint pain, headache, and tiredness recorded on the VRC. 3. **Number of Participants With Immediate AEs Following Vaccination.**

7.2 Secondary & Exploratory Outcomes * Immunogenicity: 1. Serotype-specific Immunoglobulin G (IgG) Geometric Mean Concentrations (GMCs). 2. Change From Baseline in Serotype-specific IgG GMCs. * **Serious Adverse Events (SAEs):** Number of participants experiencing any SAEs throughout the trial duration.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Solicited AEs are predefined events participants are specifically asked about and collected via the Vaccine Report Card (VRC). Unsolicited AEs are monitored continuously via clinical evaluation. **Serious Adverse Events (SAEs):** Must be reported within 24 hours of site awareness, monitored closely due to Phase 1 study status. **Data Monitoring Committee (DMC):** Internal safety review committee established to evaluate initial tolerability and safety signals.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all participants who receive at least one dose of the assigned study vaccine). **Sample Size Justification:** $N = 126$ is adequately powered for a descriptive Phase 1 safety, tolerability, and initial immunogenicity characterization. **Handling of Missing Data:** Observed data will be analyzed; missing data for safety and tolerability endpoints will generally not be imputed.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Strict Phase 1 monitoring, including source data verification of VRC entries and blood sample chain-of-custody. **Audit Trail:** Comprehensive tracking of all electronic Case Report Forms (eCRFs) and data clarification protocols.

11. Study Identifiers & Governance

Primary ID: {{V118E-003}} **Registry Number:** {{NCT07168915}}
Posting ID: {{311987672409458874}} **Sponsor/Lead Organization:** {{Merck}} **Study Title:** {{A Study of V118E in Healthy Participants (V118E-003)}} **Official Regulatory Title:** {{A Randomized, Double-Blind Study to Evaluate the Safety, Tolerability, and Immunogenicity of V118 Formulation E in Healthy Adults}}

12. Clinical Framework (Pneumococcal Vaccine Specific)

Primary Condition / Disease Area: {{Healthy}} **Diagnosis Threshold:** {{Healthy Adult Population (Aged 18-65)}} **Condition SNOMED ID:** {{69830002 | Healthy (finding)}} **Condition HPO Code:** {{HP:0032322}} **Placebo / Control Drug:** {{Sodium Chloride (ATC Code: A12CA01)}} **Medical Methodology:** {{Active Immunization}} **Optimization Target:** {{Protective antibody response against pneumococcal serotypes}}

13. Study Procedures & Methodology

Management Pathway: {{Standard clinical pathway for Phase 1 vaccination trials}} **Education Protocol (HCP):** {{Vaccine preparation, administration, and safety monitoring training}} **Education Protocol (Patient):** {{Onsite training on identifying solicited AEs and completing the Vaccine Report Card (VRC)}} **Quality Audit Mechanism:** {{Site monitoring and VRC compliance tracking}}

14. Observation & Follow-Up Schedule

Total Duration: {{Up to 24 Months (Study span out to Nov 2027)}} **Onsite Visit Cadence:** {{Day 1, Day 29, and scheduled immunogenicity follow-ups}} **Remote Monitoring Frequency:** {{Regular Phase 1 safety calls}} **Checklist Review Interval:** {{Daily VRC review post-vaccination for the predefined reaction window}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				